

Geonung Kim

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EDUCATION

- **POSTECH, Pohang, South Korea** Mar. 2021 - Present
Ph.D Student in Artificial Intelligence
GPA: 3.82/4.30
Advisors: Dr. Sunghyun Cho, POSTECH Computer Graphics Lab.
- **Hongik University, Seoul, South Korea** Mar. 2011 - Feb. 2019
Bachelor in Architecture & Computer Science (Double Major)
GPA: 3.92/4.50

RESEARCH INTEREST

Main: Generative Modeling, World Modeling, 3D and Video Generation, Machine Learning, Deep Learning

Sub: Diffusion Model Compression, Computational Photography

EXPERIENCE

- **Spacewalk** Seoul, South Korea
Software Engineer (Full-time) Jul. 2018 - May 2020
 - **Software Development:** Developed an automatic architectural design software system
 - **Analysis of Real Estate Data:** Performed million-scale GIS data analysis and visualization
- **Hyunjoon Yoo Architects** Seoul, South Korea
3D Modeler & Graphic Designer (Intern) Jul. 2015 - Aug. 2015
 - **3D Modeling:** Performed large-scale 3D building modeling and rendering using Rhino and SketchUp

PUBLICATIONS

- VideoFrom3D: 3D Scene Video Generation via Complementary Image and Video Diffusion Models
G. Kim, J. Han, S. Cho (SIGGRAPH-Asia 2025)
- LayeringDiff: Layered Image Synthesis via Generation, then Disassembly with Generative Knowledge
K. Kang, G. Sim, **G. Kim**, D. Kim, S. Nam, S. Cho (ArXiv 2025)
- DC-VSR: Spatially and Temporally Consistent Video Super-Resolution with Video Diffusion Prior
J. Han, G. Sim, **G. Kim**, HS. Lee, K. Choi, Y. Han, S. Cho (SIGGRAPH 2025)
- Diffusion Model Compression for Image-to-Image Translation
G. Kim, B. Kim, E. Park, S. Cho (ACCV 2024)
- RNA: Video Editing with ROI-based Neural Atlas
J. Lee*, **G. Kim***, S. Cho (ACCV 2024); * equal contribution
- 360 Reconstruction from a Single Image Using Space Carved Outpainting
N. Ryu, M. Gong, **G. Kim**, JH. Lee, S. Cho (SIGGRAPH-Asia 2023)
- BigColor: Colorization using a Generative Color Prior for Natural Images
G. Kim, K. Kang, S. Kim, H. Lee, S. Kim, J. Kim, SH. Baek, S. Cho (ECCV 2022)
- Realistic Blur Synthesis for Learning Image Deblurring
J. Rim, **G. Kim**, J. Kim, J. Lee, S. Lee, S. Cho (ECCV 2022)
- DynaGAN: Dynamic Few-shot Adaptation of GANs to Multiple Domains
S. Kim, K. Kang, **G. Kim**, SH. Baek, S. Cho (SIGGRAPH-Asia 2022)
- Dr. 3d: Adapting 3D GANs to Artistic Drawings
W. Jin, N. Ryu, **G. Kim**, SH. Baek, S. Cho (SIGGRAPH-Asia 2022)

PEER REVIEW ACTIVITIES

- Association for the Advancement of Artificial Intelligence (AAAI) 2026
- British Machine Vision Conference (BMVC) 2025
- Computer & Graphics 2025
- International Journal of Computer Vision (IJCV) 2023

HONORS AND AWARDS

- Outstanding Paper Award, POSTECH Graduate School of Artificial Intelligence (Feb. 2023)

COMPUTER SKILLS

- Languages: Python, C#, Bash, Lua, C/C++, SQL
- Frameworks: Pytorch, TensorFlow, .NET Framework
- Tools: Git, Docker, PostgreSQL, PostGIS, VIM, Blender, Rhino, SketchUp, 3D-Max, Photoshop